

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1513

COURSE OUTCOME COVERED

CO2: Analyse virtualisation architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

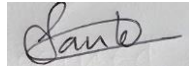
Total Marks: 25

Code of Conduct

I, SANDEEPK / 192521253(Name / Reg No), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

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CSA1513 – Cloud Computing and Big Data Analytics | Assessment Tool 3: Application Based Questions (CO2)

Question 1: Auto-Scaling and Load Balancing for Seasonal Traffic Spikes

Auto-Scaling and Load Balancing Mechanism

Cloud auto-scaling automatically increases or decreases the number of active virtual machine instances or containers based on real-time demand. During a seasonal sale, when incoming traffic rises sharply, the auto-scaling policy monitors metrics such as CPU utilization, memory usage, and request count, and provisions additional instances the moment predefined thresholds are crossed. Load balancing works alongside this by distributing incoming user requests evenly across all available instances, using algorithms such as round robin, least connections, or weighted distribution. Together, these two mechanisms ensure that no single server becomes overwhelmed while the system scales resources up during peak periods and scales them back down once the demand subsides, keeping infrastructure costs proportional to actual usage.

Reduction in Downtime and Improved User Experience

Because new instances are added automatically before existing servers are saturated, the application continues to respond to requests without crashing or becoming unresponsive, directly reducing downtime. Load balancers also perform continuous health checks and reroute traffic away from any instance that fails or becomes slow, so a single point of failure does not disrupt the whole system. For users, this translates into consistently fast page loads, uninterrupted checkout processes, and reliable access even when thousands of shoppers are browsing simultaneously, which builds trust and reduces cart abandonment.

Role of Monitoring Tools

Monitoring tools such as cloud-native dashboards and alerting systems continuously collect metrics on latency, error rates, resource utilization, and traffic patterns. These tools trigger the auto-scaling policies by feeding real-time data into scaling rules, and they alert administrators when anomalies occur, such as a sudden error spike that scaling alone cannot fix. Monitoring also provides historical data that helps teams fine-tune scaling thresholds ahead of future sales events, making the entire system progressively more efficient and predictive rather than purely reactive.

Real-World E-Commerce Scenario

A large e-commerce platform running a Black Friday sale can see traffic surge to many times its normal volume within minutes. With auto-scaling and load balancing configured on a cloud platform, additional web and application server instances are launched automatically as visitor numbers climb, while the load balancer spreads checkout and search requests across all servers. Monitoring dashboards alert the operations team if the database layer starts to lag, allowing them to scale that tier as well. As a result, customers experience smooth browsing and checkout throughout the sale, and once the event ends, the platform scales back down automatically, avoiding unnecessary infrastructure costs for the rest of the year.

Question 2: Secure Cloud Storage for Large-Volume Backup Data

Durability and Data Redundancy

Cloud storage services achieve high durability by automatically replicating each piece of data across multiple physical devices, and often across multiple geographically separated data centers. This redundancy means that even if a disk, server, or entire facility fails, copies of the data remain intact elsewhere. Techniques such as erasure coding and cross-region replication further protect against data loss, allowing cloud providers to offer durability levels of eleven nines or higher, meaning the statistical probability of losing an object is extremely small over long periods of storage.

Encryption and Access Control

Sensitive backup data is protected using encryption both at rest and in transit. At rest, data is encrypted using strong algorithms before being written to disk, so that even if physical storage media were compromised, the data remains unreadable without the correct keys. In transit, data moving to and from the cloud is protected using secure protocols such as TLS. Access control is enforced through identity and access management policies, role-based permissions, and multifactor authentication, ensuring that only authorized personnel or systems can retrieve, modify, or delete backup files, which is critical for maintaining confidentiality and regulatory compliance.

Cost Advantages Compared to Traditional Storage

Traditional on-premises backup systems require significant upfront capital investment in storage hardware, dedicated facilities, cooling, power, and ongoing maintenance staff, much of which sits underutilized outside of peak backup windows. Cloud storage instead operates on a pay-as-you-go model, where organizations pay only for the storage capacity and data transfer they actually consume, and can choose tiered storage classes that reduce cost further for infrequently accessed archival data. This eliminates large capital expenditure, reduces maintenance overhead, and allows storage capacity to scale elastically as data volumes grow.

Disaster Recovery Use Case

Consider a company whose primary data center suffers a hardware failure or is affected by a natural disaster. Because its backup data was continuously replicated to the cloud across multiple regions, the organization can quickly restore critical databases and application data from the cloud backup to a new environment, minimizing data loss and downtime. This kind of disaster recovery setup, sometimes called backup-and-restore or pilot-light recovery, allows business operations to resume within hours rather than the days or weeks that recovery from physically damaged on-premises backups might otherwise require.

Question 3: Content Delivery Networks for Global Mobile App Deployment

CDN and Latency Reduction

A content delivery network improves application performance by caching static and frequently accessed content, such as images, videos, scripts, and app assets, on servers distributed across many geographic locations. When a user requests this content, it is served from the nearest CDN node rather than from the origin server, which may be located far away. This dramatically reduces the physical distance data must travel, cutting round-trip latency and allowing the mobile app to load faster regardless of where the user is located in the world.

Role of Edge Servers

Edge servers are the geographically distributed nodes that make up the CDN, positioned close to end users in various regions and cities. Beyond simply caching content, modern edge servers can perform computation closer to the user, such as image resizing, authentication checks, or request filtering, further reducing the load on the central origin server and the delay experienced by the user. This distributed architecture also improves resilience, since if one edge location becomes overloaded or fails, traffic can be redirected to a nearby node without disrupting service.

Cloud Provider Support for Global Availability

Major cloud providers operate their CDN and edge infrastructure across dozens of regions and hundreds of points of presence worldwide, giving even a small development team access to global reach without having to build their own international infrastructure. Providers also offer intelligent routing, such as directing users to the nearest healthy region automatically, along with global load balancing and multi-region deployment options, so the mobile app remains available and responsive even during regional outages or sudden spikes in usage from a particular part of the world.

Real-Time Streaming or Gaming Application Example

A live video streaming or online multiplayer gaming application is highly sensitive to latency, since delays cause buffering or noticeable lag during gameplay. By deploying through a CDN with edge servers positioned near major user hubs across different continents, the application can deliver video segments or game state updates from the closest possible location. This ensures that a user in Chennai and a user in New York both experience smooth, low-latency performance, which is essential for maintaining engagement and a positive real-time experience.

Question 4: Compliance and Security for a Financial Institution in the Cloud

Regulatory Compliance and Auditing

Cloud providers support regulatory compliance by maintaining certifications for major standards relevant to financial services, such as PCI DSS for payment data, ISO 27001 for information security, and regional frameworks like GDPR or local banking regulations. They provide compliance dashboards, pre-built controls, and detailed audit trails that document who accessed which resources and when. This allows financial institutions to demonstrate compliance during external audits and regulatory reviews, while shared responsibility models clarify which security controls the provider manages and which remain the customer's obligation.

Logging, Monitoring, and Encryption in Security

Comprehensive logging captures every access attempt, configuration change, and data transaction within the cloud environment, creating an immutable record that supports both security investigations and compliance reporting. Continuous monitoring analyzes these logs in real time to detect suspicious behavior, such as unauthorized access attempts or unusual data transfers, triggering automated alerts or responses. Encryption protects data both at rest and in transit using strong cryptographic standards, ensuring that even if data were intercepted or storage were breached, sensitive financial information would remain unreadable without proper decryption keys.

Ensuring Data Sovereignty Requirements

Data sovereignty requires that certain data remain within specific national or regional boundaries due to legal restrictions. Cloud providers address this by allowing customers to select specific regions for data storage and processing, guaranteeing that data does not leave the required jurisdiction. Financial institutions can further enforce sovereignty through contractual agreements, region-locking policies, and regular audits to confirm that backup copies, disaster recovery sites, and third-party integrations also comply with the same geographic restrictions.

Example Involving Sensitive Financial Transactions

A bank processing millions of daily credit card transactions can use a cloud environment configured with end-to-end encryption for transaction data, strict role-based access so that only authorized systems can read cardholder information, and detailed audit logs for every transaction record accessed. Real-time monitoring can flag anomalies, such as an unusual volume of transactions from a single account, for immediate investigation. By keeping all transaction data within the required regulatory region and maintaining continuous compliance reporting, the bank satisfies both security obligations and legal requirements while still benefiting from the scalability of the cloud.

Question 5: Containerization for Application Deployment

Containers versus Virtual Machines

Virtual machines virtualize an entire physical server, with each VM running its own full operating system on top of a hypervisor, which consumes significant memory, storage, and boot-up time. Containers, in contrast, virtualize at the operating system level, sharing the host system's kernel while packaging only the application code and its dependencies. This makes containers far lighter in weight, allowing them to start in seconds rather than minutes, consume fewer resources, and enable many more containers to run on the same hardware compared to an equivalent number of virtual machines, while still providing process-level isolation.

Role of Orchestration Tools Like Kubernetes

As the number of containers in an application grows, manually managing their deployment, scaling, and networking becomes impractical. Orchestration platforms such as Kubernetes automate this process by scheduling containers across a cluster of machines, restarting failed containers automatically, scaling the number of running instances based on demand, and managing load balancing and service discovery between them. This allows development teams to deploy complex, multi-container applications reliably and to update or roll back individual components without disrupting the entire system.

Portability Benefits Across Cloud Environments

Because a container packages the application together with all its dependencies and configuration, it behaves consistently regardless of the underlying infrastructure, whether that is a developer's laptop, an on-premises server, or any major public cloud platform. This portability eliminates the common problem of an application working in one environment but failing in another due to differing system configurations. It also allows organizations to avoid vendor lock-in, since containerized applications can be moved between cloud providers or run in hybrid and multi-cloud setups with minimal modification.

Example of a Microservices-Based Application

An online food delivery platform can be built as a set of independent microservices, such as separate services for user accounts, restaurant listings, order processing, payment handling, and delivery tracking, each packaged in its own container. Kubernetes can orchestrate these containers, automatically scaling the order-processing service during meal-time rushes while keeping other services running at normal capacity, and instantly replacing any container that crashes. This architecture allows each microservice to be developed, updated, and deployed independently, improving both development speed and overall system resilience.